Permutations

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1070 | Introductory Problems | https://cses.fi/problemset/task/1070

A permutation of integers 1,2,...,n is called *beautiful* if there are no adjacent elements whose difference is 1.

Given n, construct a beautiful permutation if such a permutation exists.

Input

The only input line contains an integer n.

Output

Print a beautiful permutation of integers 1,2,...,n. If there are several solutions, you may print any of them. If there are no solutions, print "NO SOLUTION".

Constraints

- $1 \le n \le 10^6$

Example 1

Input
5

Output
4 2 5 3 1

Example 2

Input
3

Output
NO SOLUTION